

SmartBuy

Purchasing Challenge · Market signals to decide when to buy

1. Challenge in one sentence

Core challenge

- Build a tool that helps Purchasing decide whether to buy, wait, hedge or keep monitoring key raw materials such as aluminium, PET, energy and barley, combining internal data, structured price data from Cala.ai and external sources not currently considered.

2. Business context

Purchasing decisions directly affect cost, margin, availability and supply continuity. The most relevant categories for this challenge are aluminium, PET, energy and barley. Aluminium is probably the highest-spend category at the moment, but all of them are important.

The team already monitors specialized sources such as Fastmarkets, Expana, OMIP, TTF and ICIS, along with market forecasts and their own experience. The challenge is to go further: identify signals that allow Damm to anticipate market movements before they become obvious.

The ambition is to understand what information large funds or market players might be using to move earlier, and how Damm could detect those signals earlier to make better decisions on purchase timing and hedging horizon.

3. Challenge objectives

- Detect advanced signals that anticipate upward or downward market movements.
- Recommend buy, wait, hedge or monitor.
- Indicate, where relevant, for how long it would make sense to hedge.
- Combine price data with news, regulation, geopolitics, imports, futures or speculative positions.
- Generate explainable recommendations backed by evidence.

4. Available data and external enrichment

Provided datasets	External data / methods
• Barley dataset covering the previous six months. • Structured raw material price data provided by Cala.ai. • Current reference sources: Fastmarkets, Expana, OMIP, TTF and ICIS, when available or referenced. • Teams may enrich the analysis with public or accessible external data. • For PET, vPET and rPET can be considered, with drivers such as PTA, MEG, oil, ICIS LOR, imports from Asia/Turkey and European recycling regulation.	• Teams are actively expected to search for and use additional external sources: macroeconomics, geopolitics, regulation, imports/exports, freight costs, weather, futures, COT reports, industry news or alternative data. • Cala.ai will provide structured price data and may support the identification/structuring of external sources. • Every external source must be documented: origin, date, frequency, expected reliability and how it influences the recommendation.

5. What teams are expected to build

- Recommendation system: buy, wait, hedge or monitor.
- Risk or opportunity score by raw material.
- Driver explanation: what is pushing prices up or down.
- Comparison with similar historical episodes where possible.
- Dashboard or interface where users select a raw material and see recommendation, evidence and suggested horizon.
- Code repository and real working demo.

6. Working rules

- Data provided by Damm is confidential and may only be used within the hackathon context during this weekend.
- You may and should complement Damm data with public or accessible external sources, clearly documenting each source and how it was used.
- Generative AI, LLMs, AutoML, APIs, no-code/low-code tools, notebooks, Streamlit, Power BI and other technologies are allowed.
- The solution must include a code repository and a real working demo. A static mockup or slide-only proposal is not enough.
- Use the tools you are most comfortable with, but document how to run the solution and its dependencies.

7. What will be valued

- Actionability: the solution should support a real business decision, not only display data.
- Technical robustness: the approach should be supported by analysis, models, optimization logic or clear evidence.
- Data usage: we will value cleaning, integration and enrichment of Damm data with relevant external sources.
- Explainability: recommendations should be understandable and justifiable to a business user.
- Working demo: the jury should be able to see the end-to-end flow working with real or representative data.

There will be one prize per challenge and one overall prize. The solution should be actionable, but it must also be technically sound.

8. FAQ

Do we need to cover all raw materials? If you can, it will be valued. If not, focus on one or several and justify your choice. Aluminium, PET, energy and barley are the most relevant.

Does the solution need to predict the exact price? Not necessarily. It is more useful to predict trend, risk, opportunity and an explainable recommendation.

Can it recommend volume? It would be interesting, but not mandatory. The focus is on timing, hedging, risk and explanation.

Can we mention large funds or speculative positions? Yes, as long as it is analytical and based on documented sources.

What is valued the most? An actionable, technically backed and explainable recommendation.

Can we use external data? Yes. It is encouraged. You must document the sources and how they are integrated with Damm data.

Can we use generative AI tools? Yes. You may use them for analysis, development, extraction, insight generation or conversational interfaces.

Can we take Damm data outside the hackathon? No. Damm datasets cannot be shared or reused outside the event context.

What if the data is noisy or incomplete? That is part of the challenge. Document assumptions, limitations, cleaning steps and decisions made.

What is expected at the end? A code repository and a working demo that makes the solution understandable and testable.

9. Final submission checklist

Check	Required before final submission
☐	The repository includes clear run instructions.
☐	The demo allows users to select or analyze a raw material.

☐	The solution recommends buy, wait, hedge or monitor.
☐	The recommendation includes explanation based on drivers and sources.
☐	External data is incorporated in addition to the provided datasets.
☐	Sources, assumptions and limitations are documented.
☐	The demo is functional and not only a descriptive dashboard.

Final mindset

- Build something real, executable and explainable. Prioritize a working end-to-end flow over a broad but superficial solution.